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PROTECTING PERSONALLY IDENTIFIABLE INFORMATION (PII) USING MONOCULAR DEPTH ESTIMATION

Abstract

Systems, methods, and other embodiments described herein relate to protecting personally identifiable information (PII) with the use of a monocular depth estimation. In one embodiment, a method includes acquiring an original image depicting surrounding objects present in an environment. The method includes generating a depth map from the original image using a depth model that performs monocular depth estimation. The method includes obscuring at least a portion of the original image according to the depth map to provide an obscured image. The method includes providing the obscured image.

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Background/Summary

TECHNICAL FIELD

[0001] The subject matter described herein relates, in general, to systems and methods for protecting personally identifiable information (PII) and, more particularly, to using monocular depth estimation within an image processing pipeline to obscure PII.

BACKGROUND

[0002] As technology advances, traditional concepts of security morph into more complex systems that involve additional features but also encounter additional/different risks. For example, many vehicles now include the ability to collect data about the surrounding environment, such as by using cameras to acquire images that may then be used for various functions (e.g., advanced driving assistance systems (ADAS)) of the vehicle. The ability to collect this data and use the data for the noted purposes or provide the data to another entity enables additional features for the vehicle. For example, using the data with ADAS enables improved safety within the vehicle, while communicating the data may provide for tracking vehicle maintenance, automating reminders, improving security, and so on. In the context of a manufacturer, the image data provides insights about vehicle usage, training data for improving vehicle systems, and so on.

[0003] However, the imaging data also includes personally identifiable information (PII), which is data that explicitly identifies a particular person, such as a face or license plate. Thus, with the advent of personal data privacy laws such as the California Online Privacy Protection Act (CalOPPA), the European Union General Data Protection Regulation (GDPR), and others, imaging data that is unsecured can be problematic. That is, the collection or communication of the imaging data represents a significant difficulty since the imaging data generally includes PII data that may be subject to various statutes that require secure storage. However, mechanisms for properly securing the data are generally unavailable or ill-suited for integration with existing imaging pipelines. As one example, solutions may simply filter out any image that includes PII, thereby limiting available data.

SUMMARY

[0004] Example systems and methods relate to protecting PII by using monocular depth estimation to facilitate obscuring data. As noted previously, adequately protecting PII within an imaging pipeline represents a complex task with a limited set of potential solutions. That is, protecting PII, such as faces, within image data generally involves identifying the presence of the PII and then simply removing the associated image from a data set. This results in fewer examples and holes within the data overall, thereby degrading the quality of the data for various purposes, such as training a machine learning algorithm. Accordingly, in at least one arrangement, a masking system is disclosed that implements a novel approach to protecting PII by, for example, selectively masking regions of an image with substitute information from a depth map of the same image.

[0005] For example, in one approach, the masking system is implemented as part of an image processing pipeline within a device, such as a vehicle. The masking system then initially acquires images from a camera prior to the images being communicated or stored. The masking system can then process the original image using a machine learning algorithm that functions to derive a separate representation of a scene depicted by the image. That is, in the present example, the masking system performs monocular depth estimation over the image by applying a depth model. The depth model accepts the original image and outputs a depth map that is a pixel-wise determination of depths in the scene depicted by the image. In general, the depth map is simply representing depth values and not RGB images. Thus, the representation of the scene derived by the depth model does not convey details of faces, license plates, or other PII. Instead, the depth map functions to obscure the PII while still providing a depiction of the scene from the original image.

[0006] As such, in one or more arrangements, the masking system secures the PII of the original image by providing the depth map as a substitute. Consequently, in at least one approach, the masking system discards the original image to secure the PII that is present therein while providing the depth map as a replacement. Of course, in further arrangements, the masking system may selectively obscure one or more portions of the image to protect the PII instead of wholly replacing the image with the depth map. Consider that the PII that is present within any given image is often a relatively small section of the image. Thus, discarding the entirety of the RGB information from the image and using the depth map may dispose of useful information that is not protected.

[0007] Accordingly, the masking system, in at least one arrangement, identifies PII within the image and masks portions corresponding to the PII. In one aspect, the masking system uses an additional model, such as a semantic model that performs semantic segmentation. Accordingly, the masking system processes the image using the semantic model, which identifies a semantic classification for separate pixels in the image. The semantic model may provide a fine level of detail such that identification of not just overall objects (e.g., people, vehicles, etc.) is provided but also salient portions of the objects, such as faces, license plates, and so on. Using this information along with the depth map, the system can mask the portions of the image associated with the PII alone without discarding the whole image. That is, in one example, the masking system uses information from the depth map that corresponds with portions of the image, including protected information, and masks the protected information in the image with the information from the depth without covering or otherwise discarding other areas of the image. As a result, the masking system generates a combined image that includes depth data within select areas corresponding with the PII. In this way, the masking system protects the PII while also retaining non-protected information and still providing an adequate representation of the protected portions without exploiting the PII itself, thereby improving the process of protecting the information while overcoming the noted difficulties.

[0008] In one embodiment, a masking system is disclosed. The masking system includes one or more processors and a memory communicably coupled to the one or more processors. The memory stores instructions that, when executed by the one or more processors, cause the one or more processors to acquire an original image depicting surrounding objects present in an environment. The instructions include instructions to generate a depth map from the original image using a depth model that performs monocular depth estimation. The instructions include instructions to obscure at least a portion of the original image according to the depth map to provide an obscured image. The instructions include instructions to provide the obscured image.

[0009] In one embodiment, a non-transitory computer-readable medium including instructions that, when executed by one or more processors, cause the one or more processors to perform various functions is disclosed. The instructions include instructions to acquire an original image depicting surrounding objects present in an environment. The instructions include instructions to generate a depth map from the original image using a depth model that performs monocular depth estimation. The instructions include instructions to obscure at least a portion of the original image according to the depth map to provide an obscured image. The instructions include instructions to provide the obscured image.

[0010] In one embodiment, a method is disclosed. In one embodiment, the method includes acquiring an original image depicting surrounding objects present in an environment. The method includes generating a depth map from the original image using a depth model that performs monocular depth estimation. The method includes obscuring at least a portion of the original image according to the depth map to provide an obscured image. The method includes providing the obscured image.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0011] The accompanying drawings, which are incorporated in and constitute a part of the specification, illustrate various systems, methods, and other embodiments of the disclosure. It will be appreciated that the illustrated element boundaries (e.g., boxes, groups of boxes, or other shapes) in the figures represent one embodiment of the boundaries. In some embodiments, one element may be designed as multiple elements or multiple elements may be designed as one element. In some embodiments, an element shown as an internal component of another element may be implemented as an external component and vice versa. Furthermore, elements may not be drawn to scale.

[0012] FIG. **1** illustrates one embodiment of a vehicle within which systems and methods disclosed herein may be implemented.

[0013] FIG. **2** illustrates one embodiment of a masking system that is associated with protecting PII using a depth model.

[0014] FIG. **3** illustrates one embodiment of a depth model that infers depth from a monocular image.

[0015] FIG. **4** illustrates one embodiment of an image pipeline.

[0016] FIG. **5** is a flowchart illustrating one embodiment of a method for obscuring PII within images.

DETAILED DESCRIPTION

[0017] Systems, methods, and other embodiments associated with protecting PII by using monocular depth estimation to facilitate obscuring data. As noted previously, adequately protecting PII within an imaging pipeline represents a complex task with a limited set of potential solutions. That is, protecting PII, such as faces, within image data generally involves identifying the presence of the PII and then simply removing the associated image from a data set. This results in fewer examples and holes within the data overall, thereby degrading the quality of the data for various purposes, such as training a machine learning algorithm. Accordingly, in at least one arrangement, a masking system is disclosed that implements a novel approach to protecting PII by, for example, selectively masking regions of an image with substitute information from a depth map of the same image.

[0018] For example, in one approach, the masking system is implemented as part of an image processing pipeline within a device, such as a vehicle. The masking system then initially acquires images from a camera prior to the images being communicated or stored. The masking system can then process the original image using a machine learning algorithm that functions to derive a separate representation of a scene depicted by the image. That is, in the present example, the masking system performs monocular depth estimation over the image by applying a depth model. The depth model accepts the original image and outputs a depth map that is a pixel-wise determination of depths in the scene depicted by the image. In general, the depth map is simply representing depth values and not RGB images. Thus, the representation of the scene derived by the depth model does not convey details of faces, license plates, or other PII. Instead, the depth map functions to obscure the PII while still providing a depiction of the scene from the original image.

[0019] As such, in one or more arrangements, the masking system secures the PII of the original image by providing the depth map as a substitute. Consequently, in at least one approach, the masking system discards the original image to secure the PII that is present therein while providing the depth map as a replacement. Of course, in further arrangements, the masking system may selectively obscure one or more portions of the image to protect the PII instead of wholly replacing the image with the depth map. Consider that the PII that is present within any given image is often a relatively small section of the image. Thus, discarding the entirety of the RGB information from the image and using the depth map may dispose of useful information that is not protected.

[0020] Accordingly, the masking system, in at least one arrangement, identifies PII within the image and masks portions corresponding to the PII. In one aspect, the masking system uses an additional model, such as a semantic model that performs semantic segmentation. Accordingly, the masking system processes the image using the semantic model, which identifies a semantic classification for separate pixels in the image. The semantic model may provide a fine level of detail such that identification of not just overall objects (e.g., people, vehicles, etc.) is provided but also salient portions of the objects, such as faces, license plates, and so on. Using this information along with the depth map, the system can mask the portions of the image associated with the PII alone without discarding the whole image. That is, in one example, the masking system uses information from the depth map that corresponds with portions of the image, including protected information, and masks the protected information in the image with the information from the depth without covering or otherwise discarding other areas of the image. As a result, the masking system generates a combined image that includes depth data within select areas corresponding with the PII. In this way, the masking system protects the PII while also retaining non-protected information and still providing an adequate representation of the protected portions without exploiting the PII itself, thereby improving the process of protecting the information while overcoming the noted difficulties.

[0021] Referring to FIG. **1**, an example of a vehicle **100** is illustrated. As used herein, a “vehicle” is any form of powered transport. In one or more implementations, the vehicle **100** is an automobile. While arrangements will be described herein with respect to automobiles, it will be understood that embodiments are not limited to automobiles. In some implementations, the vehicle **100** may be any electronic device (e.g., smartphone, surveillance camera, robot, etc.) that, for example, perceives an environment according to images, and thus benefits from the functionality discussed herein. In yet further embodiments, the vehicle **100** may instead be a statically mounted device, an embedded device, or another device that uses images to perceive an environment.

[0022] In any case, the vehicle **100** (or another electronic device) also includes various elements. It will be understood that, in various embodiments, it may not be necessary for the vehicle **100** to have all of the elements shown in FIG. **1**. The vehicle **100** can have a different combination of the various elements shown in FIG. **1**. Further, the vehicle **100** can have additional elements to those shown in FIG. **1**. In some arrangements, the vehicle **100** may be implemented without one or more of the elements shown in FIG. **1**. While the various elements are illustrated as being located within the vehicle **100**, it will be understood that one or more of these elements can be located external to the vehicle **100**. Further, the elements shown may be physically separated by large distances and provided as remote services (e.g., cloud-computing services, software-as-a-service (SaaS), distributed computing service, etc.).

[0023] Some of the possible elements of the vehicle **100** are shown in FIG. **1** and will be described along with subsequent figures. However, a description of many of the elements in FIG. **1** will be provided after the discussion of FIGS. **2-5** for purposes of the brevity of this description. Additionally, it will be appreciated that for simplicity and clarity of illustration, where appropriate, reference numerals have been repeated among the different figures to indicate corresponding or analogous elements. In addition, the discussion outlines numerous specific details to provide a thorough understanding of the embodiments described herein. Those of skill in the art, however, will understand that the embodiments described herein may be practiced using various combinations of these elements.

[0024] In any case, the vehicle **100** includes a masking system **170** that functions to train and implement a model to process monocular images and provide depth estimates for an environment (e.g., objects, surfaces, etc.) depicted therein. Moreover, while depicted as a standalone component, in one or more embodiments, the masking system **170** is integrated with the automated driving module **160**, the camera **126**, or another component of the vehicle **100**. The noted functions and methods will become more apparent with a further discussion of the figures.

[0025] With reference to FIG. **2**, one embodiment of the masking system **170** is further illustrated. The masking system **170** is shown as including a processor **110**. Accordingly, the processor **110** may be a part of the masking system **170** or the masking system **170** may access the processor **110** through a data bus or another communication path. In one or more embodiments, the processor **110** is an application-specific integrated circuit (ASIC) that is configured to implement functions associated with a control module **220**. In general, the processor **110** is an electronic processor, such as a microprocessor, that is capable of performing various functions, as described herein. In one embodiment, the masking system **170** includes a memory **210** that stores the control module **220**. The memory **210** is a random-access memory (RAM), read-only memory (ROM), a hard disk drive, a flash memory, or other suitable memory for storing the control module **220**. The control module **220** is, for example, computer-readable instructions that, when executed by the processor **110**, cause the processor **110** to perform the various functions disclosed herein.

[0026] Furthermore, in one embodiment, the masking system **170** includes a data store **230**. The data store **230** is, in one embodiment, an electronic data structure, such as a database, that is stored in the memory **210** or another memory, and that is configured with routines that can be executed by the processor **110** for analyzing stored data, providing stored data, organizing stored data, and so on. Thus, in one embodiment, the data store **230** stores data used by the control module **220** in executing various functions. In one embodiment, the data store **230** includes images **240**, and models **250**, which may include a depth model, and/or a semantic model, along with, for example, other information that is used by the control module **220**.

[0027] It should be appreciated that the models **250** are, for example, machine learning models. As such, the masking system **170** or another system function to train the models **250** using training data. The training data generally includes one or more monocular videos to train the depth model. The videos are comprised of a plurality of frames in the form of the images **240** that are monocular images. Of course, the images **240** may alternatively be input images for use during inference by the depth model. That is, in relation to the models **250**, it should be noted that the models **250** are first trained on a particular task (e.g., monocular depth estimation) as a pre-configuration step and then used during inference to perform the task. Accordingly, the form of the input data may vary according to the type of training as compared to inference.

[0028] In any case, as described herein, a monocular image is, for example, an image from the camera **126**, or another monocular camera, that may be part of a video, and that encompasses a field-of-view (FOV) about the vehicle **100** of at least a portion of the surrounding environment. That is, the monocular image is, in one approach, generally limited to a subregion of the surrounding environment. As such, the image may be of a forward-facing (i.e., the direction of travel) 60, 90, 120-degree FOV, a rear/side-facing FOV, or some other subregion as defined by the characteristics of the camera **126**.

[0029] The monocular image itself includes visual data of the FOV that is encoded according to a video/image standard (e.g., codec) associated with the camera **126**. In general, the characteristics of the camera **126** and a video/image standard define a format of the monocular image. Thus, while the particular characteristics can vary according to different implementations, in general, the image has a defined resolution (i.e., height and width in pixels) and format. Thus, for example, the monocular image is generally an RGB visible light image. Whichever format that the masking system **170** implements, the images **240** are monocular images in that there is no explicit additional modality indicating depth nor an explicit corresponding image from another camera from which the depth can be derived (i.e., no stereo camera pair). In contrast to a stereo image that may integrate left and right images from separate cameras mounted to generate an overlapping FOV to provide an additional depth channel, the monocular image does not include explicit depth information, such as disparity maps derived from comparing the stereo images pixel-by-pixel. Instead, the monocular

image implicitly provides depth information in the relationships of perspective and a size of elements depicted therein from which the depth model derives the depth maps.
[0030] Moreover, the monocular video may include observations of many different scenes. That is, as the camera **126** or another original source camera of the video progresses through an environment, perspectives of objects and features in the environment change, and the depicted objects/features themselves also change, thereby depicting separate scenes (i.e., particular combinations of objects/features). Thus, the masking system **170** may extract particular training sets (e.g., pairs of source and target images) of monocular images from the monocular video for training. In particular, the masking system **170** generates the sets of images from the video so that the sets of images are of the same scene are related through the depiction of the same scene. As should be appreciated, the video includes a series of monocular images that are taken in succession according to a configuration of the camera. Thus, the camera may generate the images **240** (also referred to herein as frames) of the video at regular intervals, such as every 0.033 s. That is, a shutter of the camera operates at a particular rate (i.e., frames-per-second (fps)), which may be, for example, 24 fps, 30 fps, 60 fps, etc.
[0031] For purposes of the present discussion, the fps is presumed to be 30 fps. However, it should be appreciated that the fps may vary according to a particular configuration. Moreover, the masking system **170** need not generate the images for training from successive ones (i.e., adjacent) of the video frames, but instead can generally include separate images of the same scene that are not successive as training images. Thus, in one approach, the masking system **170** selects every other image depending on the fps. In a further approach, the masking system selects every fifth image as a training pair. The greater the timing difference in the video between the images, the more pronounced a difference in camera position; however, this may also result in fewer shared features/objects between the images. As such, the pairs of training images are of a same scene and are generally constrained, in one or more embodiments, to be within a defined number of frames (e.g., 5 or fewer) to ensure correspondence of an observed scene between the monocular training images. In any case, the pairs of training images generally have the attributes of being monocular images from a monocular video that are separated by some interval of time (e.g., 0.06 s) such that a perspective of the camera changes between the pair of training images as a result of the motion of the camera through the environment while generating the video.
[0032] Moreover, while the images **240** are described as training images (i.e., for purposes of adapting the depth model to improve accuracy/understanding), the masking system **170** similarly processes images of the same/similar character after training and during inference to generate the noted outputs (i.e., the depth maps). Thus, during inference and while in use as implemented, the images **240** are instead derived from a monocular camera and may not be associated via a video. Additionally, while the depth model generates a single depth map per image, the pose model accepts inputs of multiple images (e.g., two or more) to produce outputs (i.e., a transformation between image views).
[0033] With further reference to FIG. **2**, the masking system **170** further includes the models **250**, which includes the depth model that produces the depth maps, and, in at least one approach, a semantic model. The semantic model and the depth model are, in one embodiment, machine learning algorithms. However, the particular form of the models **250** may be generally distinct. That is, for example, the depth model is a machine learning algorithm that accepts an electronic input in the form of a single monocular image and produces a depth map as a result of processing the monocular image. The exact form of the depth model may vary according to the implementation but is generally a convolutional encoder-decoder type of neural network.
[0034] As an additional explanation of one embodiment of the depth model, consider FIG. **3**. FIG. **3** illustrates a detailed view of a depth model **300**. In one embodiment, the depth model **300** has an encoder/decoder architecture. The encoder/decoder architecture generally includes a set of neural network layers, including convolutional components embodied as an encoder **310** (e.g., 2D and/or

3D convolutional layers forming an encoder) that flow into deconvolutional components embodied as a decoder **320** (e.g., 2D and/or 3D deconvolutional layers forming a decoder). In one approach, the encoder **310** accepts one of the images **240** at a time as an electronic input and processes the image to extract features therefrom. The features are, in general, aspects of the image that are indicative of spatial information that the image intrinsically encodes. As such, encoding layers that form the encoder function to, for example, fold (i.e., adapt dimensions of the feature map to retain the features) encoded features into separate channels, iteratively reducing spatial dimensions of the image while packing additional channels with information about embedded states of the features. Thus, the addition of the extra channels avoids the lossy nature of the encoding process and facilitates the preservation of more information (e.g., feature details) about the original monocular image.

[0035] Accordingly, in one embodiment, the encoder **310** is comprised of multiple encoding layers formed from a combination of two-dimensional (2D) convolutional layers, packing blocks, and residual blocks. Moreover, the separate encoding layers generate outputs in the form of encoded feature maps (also referred to as tensors), which the encoding layers provide to subsequent layers in the depth model **300**. As such, the encoder **310** includes a variety of separate layers that operate on the monocular image, and subsequently on derived/intermediate feature maps that convert the visual information of the monocular image into embedded state information in the form of encoded features of different channels.

[0036] In one embodiment, the decoder **320** unfolds (i.e., adapts dimensions of the tensor to extract the features) the previously encoded spatial information in order to derive the depth map **330** for a given image according to learned correlations associated with the encoded features. That is, the decoding layers generally function to up-sample, through sub-pixel convolutions and/or other mechanisms, the previously encoded features into the depth map **330**, which may be provided at different resolutions. In one embodiment, the decoding layers comprise unpacking blocks, two-dimensional convolutional layers, and inverse depth layers that function as output layers for different scales of the feature map. The depth map **330** is, in one embodiment, a data structure corresponding to the input image that indicates distances/depths to objects/features represented therein. Additionally, in one embodiment, the depth map **330** is a tensor with separate data values indicating depths for corresponding locations in the image on a per-pixel basis.

[0037] Moreover, the depth model **300** can further include skip connections for providing residual information between the encoder **310** and the decoder **320** to facilitate memory of higher-level features between the separate components. While a particular encoder/decoder architecture is discussed, as previously noted, the depth model **300**, in various approaches, may take different forms and generally functions to process the monocular images and provide depth maps that are per-pixel estimates about distances of objects/features depicted in the images.

[0038] Moreover, while the semantic model is not explicitly illustrated, it should be noted that the semantic model may be comprised of a similar encoder/decoder architecture that separately processes the images **240** to provide semantic mappings of the images **240**. Accordingly, the semantic model may be a convolutional neural network (CNN). In one arrangement, the semantic model and the depth model **300** may share the encoder **310** while having distinct decoder heads for the separate tasks. Additionally, the semantic model also uses images for training, but the images may be combined with semantic labels as a source of supervision to facilitate the training. In this way, the semantic model is able to learn an ontology for objects and portions of objects depicted in the images. The particular degree of identification (i.e., object, object component, etc.) may be further supplemented by, in one or more arrangements, one or more intermediate models that further segment and identify components of the overall object. In any case, the semantic model processes the images to output a semantic mapping that details classifications for objects and portions of objects depicted in the image.

[0039] As an additional note, while the models **250** are discussed as discrete units separate from the control module **220**, the models **250** are, in one or more arrangements, generally integrated, at least in part, with the control module **220**. That is, the control module **220** functions to execute various processes of the models **250** and use various data structures of the models **250** in support of such execution. Accordingly, in one embodiment, the control module **220** includes instructions that function to control the processor **110** to generate the outputs using the models **250**.

[0040] In any case, the control module **220**, in one approach, trains the depth model according to a self-supervised approach that involves generating a loss, which may include an appearance-based loss (e.g., photometric loss), a depth smoothness, and so on. However, in various arrangements, one or more of the terms may not be included or further terms may be added. Moreover, in yet further approaches, the loss calculation may not be appearance-based but may instead rely on direct comparisons of depth maps. In any case, through this training, the depth model **300** develops a learned prior of the monocular images as embodied by the internal parameters of the model **300** from the training on the images. In general, the depth model develops the learned understanding about how depth relates to various aspects of an image according to, for example, size, perspective, and so on.

[0041] It should be appreciated that the control module **220**, in one or more configurations, trains the depth model **300** and a pose model together in an iterative manner over the training data embodied by sets of images that includes a plurality of monocular images from video. Through the process of training the model **300**, the control module **220** adjusts various hyper-parameters in the depth model **300** to fine-tune the functional blocks included therein. Through this training process, the depth model **300** develops a learned prior of the monocular images as embodied by the internal parameters. In general, the depth model **300** develops the learned understanding about how depth relates to various aspects of an image according to, for example, size, perspective, and so on. Consequently, the control module **220** can provide the resulting trained depth model **300** in the masking system **170** to estimate depths from monocular images that do not include an explicit modality identifying the depths.

[0042] As a brief example of an image pipeline **400** implemented with the models **250**, consider FIG. **4**. FIG. **4** is illustrated from the perspective of hardware elements involved in acquiring the images **240** and processing the images **240** into a protected output. As shown, a camera sensor **405** initially acquires an image and passes the acquired signal to a secure analog-to-digital converter **410**. In one or more arrangements, the secure A/D converter **410** is integrated with the camera **405**. In any case, the secure A/D converter **410** may provide the image via a secure pathway to the processor **110** for analysis via the control module **220** according to the models **250**. It should be noted that the image pipeline **400** generally represents a trusted hardware architecture that may be implemented along with the described systems and methods. In general, the trusted architecture implemented via the noted components, such as the secure pathway **415** and others implement trusted mechanisms that prevent sniffing or any other copying of the data acquired and communicated therein. Thus, the PII that may be present within the images **240** as acquired via the camera sensor **405** and communicated via the image pipeline **400** into the processor **110** are secure from external interference that may result in exposing the PII. In this way, the image pipeline facilitates further securing the PII.

[0043] FIG. **5** illustrates a flowchart of a method **500** that is associated with using a depth model to secure PII. Method **500** will be discussed from the perspective of the masking system **170**. While method **500** is discussed in combination with the masking system **170**, it should be appreciated that the method **500** is not limited to being implemented within the masking system **170** but is instead one example of a system that may implement the method **500**.

[0044] At **510**, the control module **220** acquires the images **240**. In the instance of inference, the control module **220** actively acquires the images **240** from the camera **126**/**405**. That is, as one example, while the vehicle **100** is operating in an environment, the control module **220** is capturing the images **240**, which the masking system **170** can then process according to the method **500**. The images **240** are generally described as being “original images” herein. This use is in reference to an unaltered form of the image that may include PII. The images **240** depict surrounding objects present in an environment in which the camera has a particular field-of-view (FoV). Thus, the images **240** may or may not include PII.

[0045] At **520**, the control module **220** generates a depth map for the image. In one approach, the control module **220** applies the depth model to the image to derive the depth map. The depth map provides a pixel-wise estimation of depth values within a scene depicted by the image. This includes distances from the camera sensor to various objects depicted in the image, such as vehicles, buildings, road surfaces, trees, people, and so on. In general, the depth map provides depth information at a granularity of the pixel. Accordingly, the depth map includes corresponding depth values for separate pixels in the image. As such, the level of detail in the depth map relates to the resolution of the original image but is further altered in that differences in color are not generally depicted in the depth map because they do not relate to distance. As such, elements having a same depth, such as a face of a person are shown with generally similar depth values that obscure the detail of the face. Similarly, license plates generally have an overall singular depth without portraying PII that may otherwise be present. Even in the case of embossed or letters/numbering with relief, the sensitivity of the depth information is not of a degree as to expose the PII. Thus, the depth map alone generally conceals PII.

[0046] At **530**, the control module **220** obscures the original image according to the depth map to provide an obscured image. In one arrangement, the control module **220** provides no consideration of the content of the original image but simply replaces the original image with the depth map and provides the depth map as the obscured image. This further includes disposing (i.e., deleting) of the original image to secure the PII.

[0047] In further approaches, the control module **220** further analyzes the original image **240** to identify particular aspects of the image and consider how to obscure any PII that may be present. For example, in one approach, the control module **220** applies a semantic model to the original image **240** that identifies semantic classes of the surrounding objects depicted in the original image. As previously noted, the semantic classes include, for example, people, vehicles, animals, road signs, buildings, surfaces (e.g., ground, road, etc.), and so on. Moreover, beyond the general identification of classes for the objects, the semantic model, in at least one approach, identifies specific features of the objects, such as faces, license plates, house numbers, etc. As such, the control module **220** can use the semantic information in different weights to further filter and obscure any PII that may be present.

[0048] It should be appreciated that, as used herein, PII generally includes personal information, such as faces, license plates, house numbers, and so on. Of course, the masking system **170** can be adapted to include additional information, and/or different information that explicitly specified herein.

[0049] In further approaches, the control module **220** uses the semantic information provided by the semantic model to filter between images that include PII and those that do not. That is, because images without PII do not need to be protected, the control module **220** may analyze the semantic information for each image to identify whether any of the protected classes (e.g., people, vehicles, etc.) are present or not. When the protected classes are present, then the control module **220** substitutes the depth map for the original image. However, otherwise, the original image is retained and not obscured.

[0050] In yet further embodiments, in place of wholly replacing the original image with the depth map, the control module masks a portion of the original image that includes the PII. In this approach, the control module **220** further discriminates between images with protected classes and without but also identifies whether the images depict specific aspects of the semantic classes, including faces, license plates, or other particular PII. Thus, only when the PII is explicitly present will the control module **220** obscure the PII. Further, the control module **220** does not wholly replace the original image with the depth map. Instead, the control module **220** obscures the portion that is the PII. That is, the control module **220** masks the specific face(s), license plate(s), house numbers, and other PII while retaining other parts of the original image.

[0051] Because the semantic model generates a mapping on a per-pixel basis of the original image with annotations about semantic classes for the separate pixels, the control module **220** parses the annotations to identify portions that include PII and then masks only the portions. The control module **220** masks the portion of the image by, for example, replacing the portion with a corresponding portion from the depth map to remove personally identifiable information (PII). That is, the control module **220** segments a corresponding section of the depth map and replaces the same portion within the image using the segmented portion of the depth map to generate the obscured image. In this way, the control module **220** is able to retain as much of the original image as possible while still securing the PII and while still providing useful information in place of the PII.

[0052] At **540**, the control module **220** provides the obscured image. In one arrangement, the control module **220** provides the obscured image in place of the original image to secure personally identifiable information (PII). This may include, for example, deleting the original image and communicating the obscured image to other systems, such as an autonomous driving module **160** or another system remote from the masking system **170**. For example, the masking system **170** originally acquires and processes the original image into the obscured image to remove the PII, and then provides the obscured image to a remote cloud-based entity for processing to identify, for example, potholes or other conditions of a roadway. Because the PII has already been removed, there is then no concern for exposing personal information and violating any privacy laws. In a further example, the obscured image may be used within the autonomous driving module **160** to control the vehicle **100** or plan a maneuver of the vehicle **100**. However, if, for example, the vehicle encounters a manual takeover by the driver because the control was inappropriate, then the system may save the obscured image as training data to improve the module **160** for future implementation. In yet further examples, the control module **220** may simply store all of the obscured images after processing in order to generate training data for various models. In this way, the masking system **170** is able to collect potentially sensitive information and secure the information against exposure while still maintaining useful data.

[0053] FIG. **1** will now be discussed in full detail as an example environment within which the system and methods disclosed herein may operate. In some instances, the vehicle **100** is configured to switch selectively between an autonomous mode, one or more semi-autonomous operational modes, and/or a manual mode. Such switching can be implemented in a suitable manner, now known or later developed. “Manual mode” means that all of or a majority of the navigation and/or maneuvering of the vehicle is performed according to inputs received from a user (e.g., human driver). In one or more arrangements, the vehicle **100** can be a conventional vehicle that is configured to operate in only a manual mode.

[0054] In one or more embodiments, the vehicle **100** is an autonomous vehicle. As used herein, “autonomous vehicle” refers to a vehicle that operates in an autonomous mode. “Autonomous mode” refers to navigating and/or maneuvering the vehicle **100** along a travel route using one or more computing systems to control the vehicle **100** with minimal or no input from a human driver. In one or more embodiments, the vehicle **100** is highly automated or completely automated. In one embodiment, the vehicle **100** is configured with one or more semi-autonomous operational modes in which one or more computing systems perform a portion of the navigation and/or maneuvering of the vehicle along a travel route, and a vehicle operator (i.e., driver) provides inputs to the vehicle to perform a portion of the navigation and/or maneuvering of the vehicle **100** along a travel route.

[0055] The vehicle **100** can include one or more processors **110**. In one or more arrangements, the processor(s) **110** can be a main processor of the vehicle **100**. For instance, the processor(s) **110** can be an electronic control unit (ECU). The vehicle **100** can include one or more data stores **115** for storing one or more types of data. The data store **115** can include volatile and/or non-volatile memory. Examples of suitable data stores **115** include RAM (Random Access Memory), flash memory, ROM (Read Only Memory), PROM (Programmable Read-Only Memory), EPROM (Erasable Programmable Read-Only Memory), EEPROM (Electrically Erasable Programmable Read-Only Memory), registers, magnetic disks, optical disks, hard drives, or any other suitable storage medium, or any combination thereof. The data store **115** can be a component of the processor(s) **110**, or the data store **115** can be operatively connected to the processor(s) **110** for use thereby. The term "operatively connected," as used throughout this description, can include direct or indirect connections, including connections without direct physical contact.

[0056] In one or more arrangements, the one or more data stores **115** can include map data **116**. The map data **116** can include maps of one or more geographic areas. In some instances, the map data **116** can include information or data on roads, traffic control devices, road markings, structures, features, and/or landmarks in the one or more geographic areas. The map data **116** can be in any suitable form. In some instances, the map data **116** can include aerial views of an area. In some instances, the map data **116** can include ground views of an area, including 360-degree ground views. The map data **116** can include measurements, dimensions, distances, and/or information for one or more items included in the map data **116** and/or relative to other items included in the map data **116**. The map data **116** can include a digital map with information about road geometry. The map data **116** can be high quality and/or highly detailed.

[0057] In one or more arrangements, the map data **116** can include one or more terrain maps **117**. The terrain map(s) **117** can include information about the ground, terrain, roads, surfaces, and/or other features of one or more geographic areas. The terrain map(s) **117** can include elevation data in the one or more geographic areas. The map data **116** can be high quality and/or highly detailed. The terrain map(s) **117** can define one or more ground surfaces, which can include paved roads, unpaved roads, land, and other things that define a ground surface.

[0058] In one or more arrangements, the map data **116** can include one or more static obstacle maps **118**. The static obstacle map(s) **118** can include information about one or more static obstacles located within one or more geographic areas. A "static obstacle" is a physical object whose position does not change or substantially change over a period of time and/or whose size does not change or substantially change over a period of time. Examples of static obstacles include trees, buildings, curbs, fences, railings, medians, utility poles, statues, monuments, signs, benches, furniture, mailboxes, large rocks, hills. The static obstacles can be objects that extend above ground level. The one or more static obstacles included in the static obstacle map(s) **118** can have location data, size data, dimension data, material data, and/or other data associated with it. The static obstacle map(s) **118** can include measurements, dimensions, distances, and/or information for one or more static obstacles. The static obstacle map(s) **118** can be high quality and/or highly detailed. The static obstacle map(s) **118** can be updated to reflect changes within a mapped area.

[0059] The one or more data stores **115** can include sensor data **119**. In this context, "sensor data" means any information about the sensors that the vehicle **100** is equipped with, including the capabilities and other information about such sensors. As will be explained below, the vehicle **100** can include the sensor system **120**. The sensor data **119** can relate to one or more sensors of the sensor system **120**. As an example, in one or more arrangements, the sensor data **119** can include information on one or more LIDAR sensors **124** of the sensor system **120**.

[0060] In some instances, at least a portion of the map data **116** and/or the sensor data **119** can be located in one or more data stores **115** located onboard the vehicle **100**. Alternatively, or in addition, at least a portion of the map data **116** and/or the sensor data **119** can be located in one or more data stores **115** that are located remotely from the vehicle **100**.

[0061] As noted above, the vehicle **100** can include the sensor system **120**. The sensor system **120** can include one or more sensors. "Sensor" means any device, component, and/or system that can detect, and/or sense something. The one or more sensors can be configured to detect, and/or sense in real-time. As used herein, the term “real-time” means a level of processing responsiveness that a user or system senses as sufficiently immediate for a particular process or determination to be made, or that enables the processor to keep up with some external process.

[0062] In arrangements in which the sensor system **120** includes a plurality of sensors, the sensors can work independently from each other. Alternatively, two or more of the sensors can work in combination with each other. In such a case, the two or more sensors can form a sensor network. The sensor system **120** and/or the one or more sensors can be operatively connected to the processor(s) **110**, the data store(s) **115**, and/or another element of the vehicle **100** (including any of the elements shown in FIG. **1**). The sensor system **120** can acquire data of at least a portion of the external environment of the vehicle **100**.

[0063] The sensor system **120** can include any suitable type of sensor. Various examples of different types of sensors will be described herein. However, it will be understood that the embodiments are not limited to the particular sensors described. The sensor system **120** can include one or more vehicle sensors **121**. The vehicle sensor(s) **121** can detect, determine, and/or sense information about the vehicle **100** itself. In one or more arrangements, the vehicle sensor(s) **121** can be configured to detect, and/or sense position and orientation changes of the vehicle **100**, such as, for example, based on inertial acceleration. In one or more arrangements, the vehicle sensor(s) **121** can include one or more accelerometers, one or more gyroscopes, an inertial measurement unit (IMU), a dead-reckoning system, a global navigation satellite system (GNSS), a global positioning system (GPS), a navigation system **147**, and/or other suitable sensors. The vehicle sensor(s) **121** can be configured to detect, and/or sense one or more characteristics of the vehicle **100**. In one or more arrangements, the vehicle sensor(s) **121** can include a speedometer to determine a current speed of the vehicle **100**.

[0064] Alternatively, or in addition, the sensor system **120** can include one or more environment sensors **122** configured to acquire, and/or sense driving environment data. “Driving environment data” includes data or information about the external environment in which an autonomous vehicle is located or one or more portions thereof. For example, the one or more environment sensors **122** can be configured to detect, quantify and/or sense obstacles in at least a portion of the external environment of the vehicle **100** and/or information/data about such obstacles. Such obstacles may be stationary objects and/or dynamic objects. The one or more environment sensors **122** can be configured to detect, measure, quantify and/or sense other things in the external environment of the vehicle **100**, such as, for example, lane markers, signs, traffic lights, traffic signs, lane lines, crosswalks, curbs proximate the vehicle **100**, off-road objects, etc.

[0065] Various examples of sensors of the sensor system **120** will be described herein. The example sensors may be part of the one or more environment sensors **122** and/or the one or more vehicle sensors **121**. However, it will be understood that the embodiments are not limited to the particular sensors described.

[0066] As an example, in one or more arrangements, the sensor system **120** can include one or more radar sensors **123**, one or more LIDAR sensors **124** (e.g., **4** beam LiDAR), one or more sonar sensors **125**, and/or one or more cameras **126**. In one or more arrangements, the one or more cameras **126** can be high dynamic range (HDR) cameras or infrared (IR) cameras.

[0067] The vehicle **100** can include an input system **130**. An “input system” includes any device, component, system, element or arrangement or groups thereof that enable information/data to be entered into a machine. The input system **130** can receive an input from a vehicle passenger (e.g., a driver or a passenger). The vehicle **100** can include an output system **135**. An “output system” includes a device, or component, that enables information/data to be presented to a vehicle passenger (e.g., a person, a vehicle passenger, etc.).

[0068] The vehicle **100** can include one or more vehicle systems **140**. Various examples of the one or more vehicle systems **140** are shown in FIG. **1**. However, the vehicle **100** can include more, fewer, or different vehicle systems. It should be appreciated that although particular vehicle systems are separately defined, each or any of the systems or portions thereof may be otherwise combined or segregated via hardware and/or software within the vehicle **100**. The vehicle **100** can include a propulsion system **141**, a braking system **142**, a steering system **143**, throttle system **144**, a transmission system **145**, a signaling system **146**, and/or a navigation system **147**. Each of these systems can include one or more devices, components, and/or a combination thereof, now known or later developed.

[0069] The navigation system **147** can include one or more devices, applications, and/or combinations thereof, now known or later developed, configured to determine the geographic location of the vehicle **100** and/or to determine a travel route for the vehicle **100**. The navigation system **147** can include one or more mapping applications to determine a travel route for the vehicle **100**. The navigation system **147** can include a global positioning system, a local positioning system, or a geolocation system.

[0070] The processor(s) **110**, the masking system **170**, and/or the automated driving module **160** can be operatively connected to communicate with the various vehicle systems **140** and/or individual components thereof. For example, returning to FIG. **1**, the processor(s) **110** and/or the automated driving module **160** can be in communication to send and/or receive information from the various vehicle systems **140** to control the movement, speed, maneuvering, heading, direction, etc. of the vehicle **100**. The processor(s) **110**, the masking system **170**, and/or the automated driving module **160** may control some or all of these vehicle systems **140** and, thus, may be partially or fully autonomous.

[0071] The processor(s) **110**, the masking system **170**, and/or the automated driving module **160** can be operatively connected to communicate with the various vehicle systems **140** and/or individual components thereof. For example, returning to FIG. **1**, the processor(s) **110**, the masking system **170**, and/or the automated driving module **160** can be in communication to send and/or receive information from the various vehicle systems **140** to control the movement, speed, maneuvering, heading, direction, etc. of the vehicle **100**. The processor(s) **110**, the masking system **170**, and/or the automated driving module **160** may control some or all of these vehicle systems **140**.

[0072] The processor(s) **110**, the masking system **170**, and/or the automated driving module **160** may be operable to control the navigation and/or maneuvering of the vehicle **100** by controlling one or more of the vehicle systems **140** and/or components thereof. For instance, when operating in an autonomous mode, the processor(s) **110**, the masking system **170**, and/or the automated driving module **160** can control the direction and/or speed of the vehicle **100**. The processor(s) **110**, the masking system **170**, and/or the automated driving module **160** can cause the vehicle **100** to accelerate (e.g., by increasing the supply of fuel provided to the engine), decelerate (e.g., by decreasing the supply of fuel to the engine and/or by applying brakes) and/or change direction (e.g., by turning the front two wheels). As used herein, “cause” or “causing” means to make, force, compel, direct, command, instruct, and/or enable an event or action to occur or at least be in a state where such event or action may occur, either in a direct or indirect manner.

[0073] The vehicle **100** can include one or more actuators **150**. The actuators **150** can be any element or combination of elements operable to modify, adjust and/or alter one or more of the vehicle systems **140** or components thereof responsive to receiving signals or other inputs from the processor(s) **110** and/or the automated driving module **160**. Any suitable actuator can be used. For instance, the one or more actuators **150** can include motors, pneumatic actuators, hydraulic pistons, relays, solenoids, and/or piezoelectric actuators, just to name a few possibilities.

[0074] The vehicle **100** can include one or more modules, at least some of which are described herein. The modules can be implemented as computer-readable program code that, when executed by a processor **110**, implement one or more of the various processes described herein. One or more of the modules can be a component of the processor(s) **110**, or one or more of the modules can be executed on and/or distributed among other processing systems to which the processor(s) **110** is operatively connected. The modules can include instructions (e.g., program logic) executable by one or more processor(s) **110**. Alternatively, or in addition, one or more data store **115** may contain such instructions.

[0075] In one or more arrangements, one or more of the modules described herein can include artificial or computational intelligence elements, e.g., neural network, fuzzy logic, or other machine learning algorithms. Further, in one or more arrangements, one or more of the modules can be distributed among a plurality of the modules described herein. In one or more arrangements, two or more of the modules described herein can be combined into a single module.

[0076] The vehicle **100** can include one or more automated driving modules **160**. The automated driving module **160** can be configured to receive data from the sensor system **120** and/or any other type of system capable of capturing information relating to the vehicle **100** and/or the external environment of the vehicle **100**. In one or more arrangements, the automated driving module **160** can use such data to generate one or more driving scene models. The automated driving module **160** can determine a position and velocity of the vehicle **100**. The automated driving module **160** can determine the location of obstacles, obstacles, or other environmental features including traffic signs, trees, shrubs, neighboring vehicles, pedestrians, etc.

[0077] The automated driving module **160** can be configured to receive, and/or determine location information for obstacles within the external environment of the vehicle **100** for use by the processor(s) **110**, and/or one or more of the modules described herein to estimate position and orientation of the vehicle **100**, vehicle position in global coordinates based on signals from a plurality of satellites, or any other data and/or signals that could be used to determine the current state of the vehicle **100** or determine the position of the vehicle **100** with respect to its environment for use in either creating a map or determining the position of the vehicle **100** in respect to map data.

[0078] The automated driving module **160** either independently or in combination with the masking system **170** can be configured to determine travel path(s), current autonomous driving maneuvers for the vehicle **100**, future autonomous driving maneuvers and/or modifications to current autonomous driving maneuvers based on data acquired by the sensor system **120**, driving scene models, and/or data from any other suitable source. “Driving maneuver” means one or more actions that affect the movement of a vehicle. Examples of driving maneuvers include: accelerating, decelerating, braking, turning, moving in a lateral direction of the vehicle **100**, changing travel lanes, merging into a travel lane, and/or reversing, just to name a few possibilities. The automated driving module **160** can be configured to implement determined driving maneuvers. The automated driving module **160** can cause, directly or indirectly, such autonomous driving maneuvers to be implemented. As used herein, “cause” or “causing” means to make, command, instruct, and/or enable an event or action to occur or at least be in a state where such event or action may occur, either in a direct or indirect manner. The automated driving module **160** can be configured to execute various vehicle functions and/or to transmit data to, receive data from, interact with, and/or control the vehicle **100** or one or more systems thereof (e.g., one or more of vehicle systems **140**).

[0079] Detailed embodiments are disclosed herein. However, it is to be understood that the disclosed embodiments are intended only as examples. Therefore, specific structural and functional details disclosed herein are not to be interpreted as limiting, but merely as a basis for the claims and as a representative basis for teaching one skilled in the art to variously employ the aspects herein in virtually any appropriately detailed structure. Further, the terms and phrases used herein are not intended to be limiting but rather to provide an understandable description of possible implementations. Various embodiments are shown in FIGS. **1-5**, but the embodiments are not limited to the illustrated structure or application.

[0080] The flowcharts and block diagrams in the Figures illustrate the architecture, functionality, and operation of possible implementations of systems, methods, and computer program products according to various embodiments. In this regard, each block in the flowcharts or block diagrams may represent a module, segment, or portion of code, which comprises one or more executable instructions for implementing the specified logical function(s). It should also be noted that, in some alternative implementations, the functions noted in the block may occur out of the order noted in the figures. For example, two blocks shown in succession may, in fact, be executed substantially concurrently, or the blocks may sometimes be executed in the reverse order, depending upon the functionality involved.

[0081] The systems, components and/or processes described above can be realized in hardware or a combination of hardware and software and can be realized in a centralized fashion in one processing system or in a distributed fashion where different elements are spread across several interconnected processing systems. Any kind of processing system or another apparatus adapted for carrying out the methods described herein is suited. A typical combination of hardware and software can be a processing system with computer-usable program code that, when being loaded and executed, controls the processing system such that it carries out the methods described herein. The systems, components and/or processes also can be embedded in a computer-readable storage, such as a computer program product or other data programs storage device, readable by a machine, tangibly embodying a program of instructions executable by the machine to perform methods and processes described herein. These elements also can be embedded in an application product which comprises all the features enabling the implementation of the methods described herein and, which when loaded in a processing system, is able to carry out these methods.

[0082] Furthermore, arrangements described herein may take the form of a computer program product embodied in one or more computer-readable media having computer-readable program code embodied, e.g., stored, thereon. Any combination of one or more computer-readable media may be utilized. The computer-readable medium may be a computer-readable signal medium or a computer-readable storage medium. The phrase “computer-readable storage medium” means a non-transitory storage medium. A computer-readable storage medium may be, for example, but not limited to, an electronic, magnetic, optical, electromagnetic, infrared, or semiconductor system, apparatus, or device, or any suitable combination of the foregoing. More specific examples (a non-exhaustive list) of the computer-readable storage medium would include the following: a portable computer diskette, a hard disk drive (HDD), a solid-state drive (SSD), a read-only memory (ROM), an erasable programmable read-only memory (EPROM or Flash memory), a portable compact disc read-only memory (CD-ROM), a digital versatile disc (DVD), an optical storage device, a magnetic storage device, or any suitable combination of the foregoing. In the context of this document, a computer-readable storage medium may be any tangible medium that can contain, or store a program for use by or in connection with an instruction execution system, apparatus, or device.

[0083] Generally, module, as used herein, includes routines, programs, objects, components, data structures, and so on that perform particular tasks or implement particular data types. In further aspects, a memory generally stores the noted modules. The memory associated with a module may be a buffer or cache embedded within a processor, a RAM, a ROM, a flash memory, or another suitable electronic storage medium. In still further aspects, a module as envisioned by the present disclosure is implemented as an application-specific integrated circuit (ASIC), a hardware component of a system on a chip (SoC), as a programmable logic array (PLA), or as another suitable hardware component that is embedded with a defined configuration set (e.g., instructions) for performing the disclosed functions.

[0084] Program code embodied on a computer-readable medium may be transmitted using any appropriate medium, including but not limited to wireless, wireline, optical fiber, cable, RF, etc., or any suitable combination of the foregoing. Computer program code for carrying out operations for aspects of the present arrangements may be written in any combination of one or more programming languages, including an object-oriented programming language such as Java™, Smalltalk, C++ or the like and conventional procedural programming languages, such as the “C”

programming language or similar programming languages. The program code may execute entirely on the user's computer, partly on the user's computer, as a stand-alone software package, partly on the user's computer and partly on a remote computer, or entirely on the remote computer or server. In the latter scenario, the remote computer may be connected to the user's computer through any type of network, including a local area network (LAN) or a wide area network (WAN), or the connection may be made to an external computer (for example, through the Internet using an Internet Service Provider).
[0085] The terms “a” and “an,” as used herein, are defined as one or more than one. The term “plurality,” as used herein, is defined as two or more than two. The term “another,” as used herein, is defined as at least a second or more. The terms “including” and/or “having,” as used herein, are defined as comprising (i.e., open language). The phrase “at least one of . . . and . . . ” as used herein refers to and encompasses any and all possible combinations of one or more of the associated listed items. As an example, the phrase “at least one of A, B, and C” includes A only, B only, C only, or any combination thereof (e.g., AB, AC, BC or ABC).
[0086] Aspects herein can be embodied in other forms without departing from the spirit or essential attributes thereof. Accordingly, reference should be made to the following claims, rather than to the foregoing specification, as indicating the scope hereof.

Claims

1. A masking system, comprising: one or more processors; a memory communicably coupled to the one or more processors and storing instructions that, when executed by the one or more processors, cause the one or more processors to: acquire an original image depicting surrounding objects present in an environment; generate a depth map from the original image using a depth model that performs monocular depth estimation; obscure at least a portion of the original image according to the depth map to provide an obscured image; and provide the obscured image.
2. The masking system of claim 1, wherein the depth model performs monocular depth estimation and is trained according to self-supervised structure-from-motion (SfM) training.
3. The masking system of claim 1, wherein the instructions include instructions to obscure the original image include instructions to replace the original image with the depth map and disposing of the original image to secure personally identifiable information (PII) in the original image.
4. The masking system of claim 1, wherein the instructions include instructions to obscure the original image include instructions to apply a semantic model to the original image that identifies semantic classes of the surrounding objects depicted in the original image, including at least people.
5. The masking system of claim 4, wherein the instructions include instructions to obscure the original image include instructions to mask at least a portion of the surrounding objects by replacing the portion of the surrounding objects with corresponding portions from the depth map to remove personally identifiable information (PII).
6. The masking system of claim 5, wherein the instructions include instructions to mask include instructions to place the portions of the depth map to obscure one or more of: faces and license plates.
7. The masking system of claim 1, wherein the instructions include instructions to provide the obscured image include instructions to provide the obscured image in place of the original image to secure personally identifiable information (PII).
8. The masking system of claim 1, wherein the masking system is integrated into an image processing pipeline within a vehicle to secure personally identifiable information (PII) in the original image during operation of the vehicle.
9. A non-transitory computer-readable medium including instructions that, when executed by one or more processors, cause the one or more processors to: acquire an original image depicting

surrounding objects present in an environment; generate a depth map from the original image using a depth model that performs monocular depth estimation; obscure at least a portion of the original image according to the depth map to provide an obscured image; and provide the obscured image.

10. The non-transitory computer-readable medium of claim 9, wherein the depth model performs monocular depth estimation and is trained according to self-supervised structure-from-motion (SfM) training.

11. The non-transitory computer-readable medium of claim 9, wherein the instructions include instructions to obscure the original image include instructions to replace the original image with the depth map and disposing of the original image to secure personally identifiable information (PII) in the original image.

12. The non-transitory computer-readable medium of claim 9, wherein the instructions include instructions to obscure the original image include instructions to apply a semantic model to the original image that identifies semantic classes of the surrounding objects depicted in the original image, including at least people.

13. The non-transitory computer-readable medium of claim 12, wherein the instructions include instructions to obscure the original image include instructions to mask at least a portion of the surrounding objects by replacing the portion of the surrounding objects with corresponding portions from the depth map to remove personally identifiable information (PII).

14. A method, comprising: acquiring an original image depicting surrounding objects present in an environment; generating a depth map from the original image using a depth model that performs monocular depth estimation; obscuring at least a portion of the original image according to the depth map to provide an obscured image; and providing the obscured image.

15. The method of claim 14, wherein the depth model performs monocular depth estimation and is trained according to self-supervised structure-from-motion (SfM) training.

16. The method of claim 14, wherein obscuring the original image includes replacing the original image with the depth map and disposing of the original image to secure personally identifiable information (PII) in the original image.

17. The method of claim 14, wherein obscuring the original image includes applying a semantic model to the original image that identifies semantic classes of the surrounding objects depicted in the original image, including at least people.

18. The method of claim 17, wherein obscuring the original image includes masking at least a portion of the surrounding objects by replacing the portion of the surrounding objects with corresponding portions from the depth map to remove personally identifiable information (PII).

19. The method of claim 18, wherein masking includes placing the portions of the depth map to obscure one or more of: faces and license plates.

20. The method of claim 14, wherein providing the obscured image includes providing the obscured image in place of the original image to secure personally identifiable information (PII).